

Determining critical phase transition temperature in an Ising model simulation

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1 Abstract

TODO: <Write abstract>

2 Introduction

2.1 Aim of study

In this article we look at determining at what temperature does the Ising model undergo a phase transition.

2.2 The Ising model

TODO: <Description of the Ising model>

3 Methodology

3.1 Metropolis algorithm

TODO: <Explain metropolis algorithm>

3.2 Determining parameters for final simulation

TODO: <Explain how we arrived at simulation parameters>

4 Results

TODO: <Give results>

TODO: <Give figures of results>

5 Comparison to theory

5.1 Theoretical solution for the critical phase transition temperature

TODO: <Give derivation of critical temperature in the Ising model>

5.2 Comparison of theory and our results

TODO: <Compare our results with theoreticla values>

6 Conclusion

TODO: <Conclude that our simulated values are close to theoretical vlaues>

TODO: <Give potential improvements on methodology>

7 Appendix

TODO: <Paste code>